

FINAL PROJECT REPORT

AI-Assisted Marine Risk Intelligence & Community Reporting Platform

A geospatial marine intelligence system that fuses satellite sea-surface temperature data, community-reported coastal incidents, and simulation tools to identify at-risk marine zones and support early response.

Prepared by Team of 8

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Project Duration: 15 Working Days

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Table of Contents

1. Executive Summary
2. Project Objectives
3. Team Structure, Role Allocation & Coordination
 - 3.1 Role Allocation Table
 - 3.2 Student Identification Details
 - 3.3 Team Coordination & Acknowledgement
4. System Architecture & Technology Stack
5. Algorithm Design
6. Day 1 – Project Initiation Summary
7. Individual Member Progress Reports (Day-wise, 15 Days)
 - 7.1 Deepankar Singh Sirari – Team Lead, Documentation & UI/Frontend Co-Design
 - 7.2 Navneet Kumar – Data Collection, Cleaning & GIS Pipeline
 - 7.3 Suhana Singh – Data Cleaning & Sorting
 - 7.4 Ayush Kumar – GIS Engineer
 - 7.5 Shivansh Sharma – Database Engineer
 - 7.6 Sumit Sharma – Backend Developer & Deployment
 - 7.7 Aditya Pratap Singh – Frontend Developer
 - 7.8 Lakshya Narula – Deployment & Moderation, Hardware Research & Algorithm Support
8. Testing & Quality Assurance Summary
9. Final Deliverables
10. Challenges & Solutions
11. Conclusion
12. References



1. Executive Summary

This report documents the complete work carried out by an eight-member student team over a fifteen-working-day project period to design and build an AI-assisted marine risk intelligence and community reporting platform. The system fuses satellite-derived Sea Surface Temperature (SST) data, historical ocean chemistry parameters, community-reported coastal incidents, and simulation tools to identify at-risk marine zones and support early environmental response.

The project began as a concept for an AI-based water quality monitoring system using low-cost sensors (ESP32, pH, turbidity, TDS) for NGO use. During Day 1 planning, the team refined this into a broader geospatial marine intelligence platform built on satellite SST data, GIS visualization, community reporting, and risk simulation. As the physical ESP32 microcontroller and sensor hardware were not procured or made available during the project period, the team adapted its scope to deliver a fully software- and satellite-data-driven platform, documenting the sensor-based extension as a future-phase recommendation rather than a built component.

Each team member was assigned a focused responsibility — data collection and cleaning, GIS engineering, backend development and deployment, database engineering, frontend and UI design, algorithm design, and documentation — and maintained a day-wise log of their contribution across all fifteen days, compiled in Section 7 of this report. The first ten days were spent on research, data processing, GIS correction, backend/database foundation, and interface design; the final five days (Days 11–15) were dedicated to integration, deployment, testing, refinement, and final demo preparation.

Over the fifteen working days, the team produced a cleaned and analysis-ready SST and ocean chemistry dataset, a working SST anomaly calculation and GIS visualization pipeline, a deployed PostgreSQL/PostGIS database with an automated ingestion pipeline, a deployed backend API with endpoints for zone risk, alerts and simulation, a fully integrated frontend dashboard with GIS map, alerts and community reporting modules, a complete algorithm design with pseudocode, and a full testing and quality-assurance module validating the pipeline end to end. Every member contributed a distinct, essential piece of this outcome, and the team coordinated closely throughout — through daily updates, shared planning, and cross-functional collaboration — to deliver a complete, working platform.

2. Project Objectives

The platform is designed to:

- Ingest satellite sea surface temperature (SST) data from trusted sources (NOAA, OISST, NCEI, OceanSODA).
- Compute SST anomalies against a historical (1991–2020) climatology baseline to detect abnormal warming events.
- Display GIS layers for SST, anomalies, and marine risk zones.

- Allow users to simulate the spread or impact of a heatwave event.
- Allow community reporting of pollution or harmed coastal areas, with verification against risk zones.
- Summarize marine-health impact through an interactive, deployed dashboard to support decision-making by NGOs and coastal authorities.

The project stands out because it combines satellite science, web GIS, public/community reporting, marine impact analysis, and simulation into a single workflow – going beyond a conventional dashboard to address a real environmental monitoring problem with a map-based decision support tool.

3. Team Structure, Role Allocation & Coordination

3.1 Role Allocation Table

The team of eight divided responsibilities as follows. Roles were finalized on Day 1 and refined as the project progressed to a fifteen-day timeline.

#	Member	Primary Role	Key Responsibilities
1	Navneet Kumar	Data Collector & Handler	Collects SST/ocean chemistry datasets, cleans them, prepares GIS-ready inputs
2	Suhana Singh	Data Cleaner & Sorter	Supports data collection, cleaning, and quality checks
3	Ayush Kumar	GIS Engineer	Builds maps, layers, markers, coastal overlays, spatial corrections
4	Sumit Sharma	Backend Developer & Deployment	API endpoints, data ingestion, alerts, simulation services, deployment
5	Shivansh Sharma	Database Engineer	PostGIS/geospatial schema, indexing, data pipeline automation, deployment support
6	Aditya Pratap Singh	Frontend Developer	Dashboard pages, responsive layout, charts & widgets
7	Lakshya Narula	Deployment & Moderation / Hardware Research / Algorithm Support	Community reporting forms, admin review flow, hardware research, algorithm & presentation support
8	Deepankar Singh Sirari	Team Lead / Documentation Lead / UI & Frontend Co-Design	Report writing, PPT, algorithm design, UI/frontend design collaboration, deployment & demo coordination

3.2 Student Identification Details

Member	Roll No.	Aadhar No.
Deepankar Singh Sirari	25SCS1003002918	263074521328
Navneet Kumar	25SCS1003003086	703088434588



Member	Roll No.	Aadhar No.
Suhana Singh	25SCS1003000744	949912711994
Ayush Kumar	25SCS1003003078	601745257567
Shivansh Sharma	25SCS1003004651	263943997350
Sumit Sharma	25SCS1003002486	898424069533
Aditya Pratap Singh	25SCS1003000571	806024918757
Lakshya Narula	25SCS1003000518	688239078650

3.3 Team Coordination & Acknowledgement

Every member of this eight-person team contributed a distinct and essential piece of the final platform, and each gave their best effort across all fifteen working days – from early-stage research and data handling to backend, database, GIS, frontend, deployment, and documentation work. The team coordinated closely throughout the project: role responsibilities were agreed upfront on Day 1, progress was shared and reviewed regularly, and cross-functional pairs (data ↔ GIS, backend ↔ database, frontend ↔ UI design) worked together at key integration points so that individual contributions came together into a single, coherent platform rather than disconnected pieces.

This section records that appreciation formally: the data team (Navneet and Suhana) ensured every downstream module worked with clean, trustworthy data; the GIS team (Ayush) translated that data into meaningful maps; the database and backend team (Shivansh and Sumit) built and deployed the system that ties it all together; the frontend and UI team (Aditya and Deepankar) made the platform usable and presentable; Lakshya supported the moderation workflow, hardware research, and presentation preparation; and the team lead (Deepankar) held the documentation, algorithm design, and overall coordination together from Day 1 through final submission. The completeness of this report – and of the platform it documents – reflects the consistent, well-coordinated effort of all eight members.

4. System Architecture & Technology Stack

4.1 Final Deliverables Targeted

- Interactive, deployed dashboard
- GIS map with layers
- SST anomaly visualizer
- Community reporting module
- Simulation module
- Admin moderation panel
- Backend API + database (deployed)



- Final report + PPT + demo video

4.2 Technology Stack

Layer	Technologies Used
Data Processing	Python (Pandas, NumPy, Xarray), NetCDF, CSV
Geospatial Analysis	QGIS (EPSG:4326 / WGS 84), IDW interpolation, PostGIS
Database	PostgreSQL, pgAdmin, PostGIS, Prisma ORM (deployed)
Backend	Node.js, Python-to-PostgreSQL automation pipeline, REST API (deployed)
Frontend / Dashboard	Streamlit / dashboard components, Plotly Express, responsive web design
Data Sources	NOAA, OISST, NCEI, OceanSODA

4.3 Hardware Scope

At project initiation, an ESP32 microcontroller with pH, Turbidity, TDS, and Temperature sensors was planned as a future physical-sensing extension of the platform, to be programmed via Arduino IDE.

***Note:** The ESP32 microcontroller and the associated sensor kit were not procured or made available during the fifteen-day project period, and Arduino-based firmware development was consequently not carried out. The team proceeded with a fully software- and satellite-data-based implementation for this phase. Hardware integration remains a documented, ready-to-build recommendation for a future phase, based on the research completed by Lakshya Narula (see Section 7.8).*

5. Algorithm Design

5.1 Natural Language Description

The algorithm collects SST data and community reports, validates the inputs, computes SST anomalies, classifies marine risk, verifies reports, generates alerts, runs simulations, and updates the dashboard.

5.2 Input

- Satellite SST observations
- Climatology baseline values
- Coastal zone polygons
- Community reports
- Simulation parameters

5.3 Output

- SST anomaly values
- Heatwave flags

- Zone risk levels
- Verified reports
- Alerts
- Simulation polygons
- Updated dashboard data

5.4 Definitions

- SST anomaly: Difference between observed SST and climatology.
- Heatwave flag: Indicator that SST exceeds the risk threshold.
- Zone risk score: Weighted score based on anomaly, cell intensity, and reports.
- Verified report: A report matching a risky geographic area.
- Simulation output: Predicted spread or impact area under chosen parameters.

5.5 Finiteness

The algorithm stops after processing all SST records, all zones, all reports, and the simulation request. Every loop is bounded by the size of the input data. Therefore, the algorithm is finite.

5.6 Effectiveness

Every step uses basic operations such as reading, validating, comparing, counting, averaging, and storing. Each step is executable in a finite number of actions. Therefore, the algorithm is effective.

5.7 Algorithm Properties

Correctness: Produces accurate anomaly values, correct zone classification, valid report verification, and proper alerts.

Efficiency: Uses structured steps for SST processing, geospatial lookup, and dashboard update.

Scalability: Can handle more SST records, more zones, and more reports by processing them in batches or through spatial indexing.

Simplicity: Uses clear rules for anomaly thresholds, risk scoring, and verification.

Maintainability: Each part is separated into modules: ingestion, map rendering, reporting, simulation, alerts, and database.

Robustness: Handles missing values, invalid reports, and failed API calls without breaking the system.

5.8 Pseudocode

```
BEGIN
  Collect SST data, reports, and simulation inputs
  FOR each SST record
    Validate record
```



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```
    Compute anomaly
    Classify risk
    Store record
END FOR
FOR each coastal zone
    Aggregate SST cells
    Compute risk score
    Assign zone risk level
END FOR
FOR each community report
    Validate report
    Compare with risky zones
    Mark verified or pending
END FOR
IF zone risk is high or critical THEN
    Generate alert
END IF
Run simulation
Update dashboard
END
```

5.9 System Modules

- Data collection module
- SST processing module
- Geospatial zone analysis module
- Community report module
- Alert generation module
- Simulation module
- Frontend dashboard module
- Backend API module
- Database storage module



6. Day 1 – Project Initiation Summary

Day 1 focused on problem identification, scope finalization, technology selection, role allocation, and roadmap planning for the following 15 working days.

Problem Identification

The team discussed the growing issue of water pollution and the limitations of traditional manual testing methods, and concluded that a low-cost, AI-enabled monitoring system could help NGOs continuously assess water quality and respond quickly to contamination. This was subsequently broadened into a geospatial marine intelligence system fusing satellite SST data, community-reported coastal incidents, and simulation tools.

Project Scope

The team finalized the concept of building a web platform that ingests satellite sea surface temperature data, shows GIS layers for SST, anomalies and risk zones, lets users simulate spread or heatwave impact, allows community reporting of pollution or harmed areas, and summarizes marine-health impact for decision-making.

Technology Selection (Initial Plan)

- Microcontroller: ESP32 (planned; not procured – see Section 4.3)
- Sensors: pH, Turbidity, TDS, Temperature (planned; not procured – see Section 4.3)
- Programming: Arduino IDE (ESP32 firmware, not undertaken) and Python (AI/data module)
- Database: Firebase Realtime Database (initial idea) → PostgreSQL/PostGIS (final, deployed architecture)
- Dashboard: Streamlit / web dashboard

Work Initiated on Day 1

- Identified the project problem and objectives.
- Finalized the overall system architecture.
- Assigned roles and responsibilities to all team members.
- Selected the required technologies and software stack.
- Drafted the implementation roadmap and estimated resource requirements.
- Planned a 15-day implementation schedule covering data collection, AI/data development, dashboard creation, testing, deployment, and final presentation.

Outcome of Day 1

By the end of Day 1, the team established a clear project direction, assigned responsibilities, selected the technology stack, and developed a structured 15-day execution plan for building the AI-powered marine monitoring platform.

7. Individual Member Progress Reports (Day-wise, 15 Days)

Each member's fifteen-day log below reflects the two natural phases of the project: Days 1–10 covered research, data/GIS work, foundational backend and database development, and initial UI/frontend design; Days 11–15 covered integration, deployment, testing, refinement, and final demo preparation.

7.1 Deepankar Singh Sirari – Team Lead, Documentation & UI/Frontend Co-Design

Deepankar Singh Sirari | *Team Lead / Documentation Lead / UI & Frontend Co-Design* | 15 Working Days
Roll No: 25SCS1003002918 | Aadhar No: 263074521328

As Team Lead and Documentation Lead, Deepankar coordinated the overall project direction, facilitated planning discussions, authored the algorithm design (Section 5), compiled the team's day-wise progress into this final report, and – alongside Aditya Pratap Singh – actively co-led the dashboard's UI and frontend design decisions, from visual direction through final styling.

Day 1: Facilitated the team kickoff discussion, led problem identification and scope finalization, and drafted the initial project roadmap and role allocation sheet.

Day 2: Documented the finalized technology stack and team responsibilities; began compiling the Day 1 progress report and set up the shared documentation structure for daily updates.

Day 3: Researched algorithmic problem-solving concepts and began drafting the natural-language description of the system algorithm; also reviewed early dashboard wireframes with Aditya and gave input on navigation structure.

Day 4: Defined the structured input and output specification for the algorithm, listing satellite SST observations, climatology baselines, coastal zone polygons, community reports, and simulation parameters as inputs, and anomaly values, risk levels, alerts, and dashboard data as outputs.

Day 5: Documented core algorithm definitions (SST anomaly, heatwave flag, zone risk score, verified report, simulation output) and analyzed the algorithm for finiteness and effectiveness.

Day 6: Evaluated the algorithm against correctness, efficiency, scalability, simplicity, maintainability, and robustness criteria, and reviewed the design with module owners for feedback.

Day 7: Authored the complete end-to-end pseudocode for the system workflow, from data collection through zone risk scoring, report verification, alert generation, simulation, and dashboard update.

Day 8: Defined the nine core system modules and mapped each to the responsible team member; collaborated with Aditya on the visual layout of the KPI cards and colour palette for the dashboard.

Day 9: Collected and reviewed day-wise progress updates from all team members, cross-checked them against the Testing Module report, and worked with Aditya to finalize the dashboard's typography and theme.

Day 10: Consolidated the ten-day progress into a working project report draft; reviewed the frontend



build end-to-end with Aditya for UI consistency ahead of the integration phase.

Day 11: Worked closely with Aditya on refining the dashboard's UI/UX for the integration phase – reviewing layout, colour palette, and typography choices to ensure consistency across the GIS map, alerts, and reporting pages.

Day 12: Reviewed and approved frontend design decisions as live backend data was integrated; contributed directly to dashboard styling, spacing, and content layout alongside Aditya.

Day 13: Coordinated the end-to-end testing schedule with the testing/QA effort and updated project documentation to reflect the extended 15-day timeline and deployment milestones.

Day 14: Consolidated final individual reports from all members covering Days 11–15, and reviewed the deployed dashboard's UI for final polish alongside Aditya.

Day 15: Finalized the complete project report, PPT, and demo video; conducted a final review of all deliverables with the team and confirmed every member's contribution was accurately represented.

Key Contribution: *Held the project's direction, documentation, and algorithm design together from Day 1 to Day 15, while also directly shaping the dashboard's UI and frontend design in close partnership with Aditya – a dual role spanning leadership and hands-on design work.*

7.2 Navneet Kumar – Data Collection, Cleaning & GIS Pipeline

Navneet Kumar | Data Collector & Handler | 15 Working Days

Roll No: 25SCS1003003086 | Aadhar No: 703088434588

Role: Research, Data Collection, Data Cleaning, Data Processing, Ocean Data Analysis, Dataset Preparation, GIS-Ready Dataset Development.

Objective: to move from theoretical research to a working, GIS-ready data pipeline – collecting reliable datasets (NOAA, OISST, NCEI, OceanSODA), preparing a historical climatology baseline, cleaning and validating the data, calculating SST anomaly, optimizing the dataset spatially, preparing ocean chemistry data, designing a multi-parameter risk framework, and documenting and optimizing the entire workflow for scalability and deployment.

Day 1: Studied NOAA resources and scientific literature on marine heatwaves, ocean warming, coral bleaching, and fish migration; learned why SST anomaly is a far better indicator of abnormal warming than raw SST, and prepared a short summary of project objectives and research direction.

Day 2: Identified and compared datasets from NOAA, OISST, NCEI, and OceanSODA by spatial/temporal resolution and grid system; downloaded SST, climatology, and ocean chemistry data in NetCDF and CSV formats and used Xarray to inspect their structure.

Day 3: Studied why the 1991–2020 baseline period is used in climate science, calculated the long-term average SST per grid location with Xarray, implemented $\text{SST Anomaly} = \text{Current SST} - \text{Historical Mean SST}$, cross-checked results against NOAA reference maps, and resolved a longitude-convention mismatch ($0-360^\circ$ vs $-180^\circ/+180^\circ$).



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Day 4: Used Pandas and NumPy to remove missing values, duplicate records, and invalid coordinates, standardized column names across sources, and cleaned incrementally with statistics checked after every step to avoid discarding genuine coastal/ice-region observations.

Day 5: Developed a Python script (Pandas, NumPy, Xarray) to align current and historical SST on shared coordinates and compute anomaly values using vectorized operations; validated results against NOAA anomaly maps and identified candidate heatwave hotspot regions.

Day 6: Reduced the high-resolution anomaly dataset to a coarser 10-degree grid by cell-wise averaging (not subsampling), preserving the global warming pattern while lowering time and space complexity and easing GIS import.

Day 7: Imported the optimized dataset into QGIS as a point layer (EPSG:4326), applied graduated NOAA -style colour-ramp symbology, experimented with IDW interpolation to produce a continuous surface, and refined classification, opacity, and legends.

Day 8: Downloaded the OceanSODA dataset, extracted pH, salinity, and surface CO₂ using the NetCDF/Xarray workflow, calculated annual mean values, removed missing data, converted longitude conventions, and renamed variables for clarity.

Day 9: Researched how SST anomaly interacts with pH, salinity, and CO₂ to compound ecological stress, and designed a workflow integrating SST anomaly with ocean chemistry into a multi-parameter risk framework with consistently structured backend datasets.

Day 10: Reviewed the complete pipeline using time/space complexity concepts, consolidated redundant processing steps, reorganized datasets into a scalable folder structure, and documented the full workflow with a summary flowchart.

Day 11: Finalized backend-ready dataset exports (SST, anomaly, and ocean chemistry tables) and coordinated the handoff with Shivansh and Sumit for database import and API integration.

Day 12: Cross-verified the dataset now integrated into the live PostgreSQL/PostGIS database against the original cleaned files to confirm no records were lost or altered during import.

Day 13: Prepared representative sample datasets of varying sizes to support the team's load and performance testing of the deployed backend and dashboard.

Day 14: Reviewed the dashboard's data visualizations (anomaly maps, trend charts) against the source dataset to confirm accuracy after full-stack integration.

Day 15: Took part in final end-to-end testing, contributed the data-methodology sections of the final report, and supported the team's demo rehearsal.

Technologies Used

Python (Pandas, NumPy, Xarray); NetCDF and CSV; QGIS (point layers, graduated symbology, colour ramps, IDW interpolation); NOAA, OISST, NCEI, OceanSODA.



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Challenges & Solutions

Key challenges included inconsistent longitude conventions, the risk of over-cleaning near coastal/missing-data regions, choosing an appropriate spatial aggregation level, achieving a clear yet accurate GIS visualization, and integrating SST anomaly with ocean chemistry. These were resolved by standardizing coordinates early with reference-point verification, cleaning incrementally with checks after every step, validating aggregated patterns against high-resolution data, iteratively testing QGIS classification/interpolation settings, and grounding the risk framework only in relationships supported by scientific literature.

Key Contribution: *Built the foundational, analysis-ready SST and ocean-chemistry dataset that every other module in the platform – GIS, database, backend, and dashboard – ultimately depended on.*

7.3 Suhana Singh – Data Cleaning & Sorting

Suhana Singh | Data Cleaner & Sorter | 15 Working Days

Roll No: 25SCS1003000744 | Aadhar No: 949912711994

Role: Research Support, Data Collection Assistance, Data Cleaning & Data Analysis Support, working alongside Navneet Kumar who led the core data processing.

Day 1: Read introductory material on marine heatwaves, ocean warming, and coral bleaching; learned the basic idea behind SST anomaly and discussed the overall fifteen-day plan with the team.

Day 2: Was introduced to the NOAA, OISST, NCEI, and OceanSODA data sources and helped verify that the correct time period and region had been downloaded before further processing.

Day 3: Learned why a fixed 1991–2020 baseline is used instead of a single earlier year, sat with the SST anomaly calculation, and helped spot-check early results against reference values.

Day 4: Began supporting data cleaning by checking for missing values and duplicate rows, and flagged rows with out-of-range coordinates for review.

Day 5: Checked latitude/longitude consistency across files and helped compare and standardize column names between datasets, keeping a record of what was removed at each step.

Day 6: Learned that coastal and ice-covered areas need extra care during cleaning, and helped re-check flagged rows in these areas to confirm genuine readings versus real errors.

Day 7: Reviewed the cleaned dataset with the team and helped compare row counts and value ranges before/after cleaning to confirm the cleaning steps worked as expected.

Day 8: Was introduced to positive vs negative SST anomalies and helped identify regions where anomaly values stood out, feeding into the team's discussion of possible warming hotspots.

Day 9: Assisted in organizing the cleaned data into a consistent format, double-checked column names and units across files, and helped label and file the datasets for the team.

Day 10: Reviewed the first ten days of work with the team, helped note down the key steps taken during



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collection, cleaning, and initial analysis, and prepared for the integration phase.

Day 11: Helped re-verify final dataset consistency after the data was imported into the live database, checking a sample of records against the original files.

Day 12: Assisted in preparing sample data subsets used for dashboard demo scenarios, ensuring a good mix of low, moderate, and high anomaly regions.

Day 13: Supported data-related test cases during the team's testing phase, helping confirm that missing values and duplicate records were correctly handled end to end.

Day 14: Helped review the data displayed on the deployed dashboard for accuracy against the cleaned source files.

Day 15: Took part in the final review meeting and helped compile the data-cleaning learnings and outcomes for the final report.

Technologies & Skills Learned

Python (Pandas) for tabular ocean data; basic understanding of CSV and NetCDF formats; how research on a real environmental problem becomes a data project; careful, incremental data cleaning; and working as part of a small collaborative team.

Challenges & Solutions

The main challenge was distinguishing genuine errors from valid but uncommon ocean readings, particularly near coastlines — addressed by checking flagged rows carefully with the team rather than removing data unilaterally, and asking questions whenever a result did not immediately make sense.

Key Contribution: *Provided steady, careful support across the entire data-cleaning process and through final verification — the quiet, consistent quality checks that kept the dataset trustworthy from Day 1 through deployment.*

7.4 Ayush Kumar – GIS Engineer

Ayush Kumar | GIS Engineer | 15 Working Days

Roll No: 25SCS1003003078 | Aadhar No: 601745257567

Role: Builds maps, layers, markers, and coastal overlays for the Marine Heatwave Geospatial Mapping & Dashboard Pipeline.

Days 1–3: Familiarized with the structure of the raw marine heatwave dataset; identified a critical geospatial indexing error where satellite telemetry recorded longitude on a 0°–360° continuous grid, breaking standard flat-map (–180° to 180°) representation in QGIS; conducted exploratory analysis on the correlation between raw SST, historical baseline SST, and thermal anomalies.

Days 4–6: Developed a coordinate transformation rule (IF Longitude > 180, Longitude – 360) to correct the 0°–360° wrap; built a Python/Pandas extraction script to apply the correction and output a clean dataset (fixed_atlantic_heatwaves.csv); applied bounding-box filters to isolate the target ecosystem



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(latitude -20° to 20° , longitude -60° to 15°).

Days 7–8: Imported the corrected dataset into QGIS using EPSG:4326 (WGS 84); resolved rendering issues such as overlapping point grids by adjusting vector layer properties; applied a Graduated Symbol Renderer on SST_Anomaly with a high-contrast sequential colour ramp styled as boundary-free square grids to isolate thermal hotspots.

Days 9–10: Designed a professional print-layout vector map in QGIS, bypassing WMS resolution limits via structured tile parameters; documented the algorithmic pipeline pseudocode; prepared backend spatial code blocks (Plotly Express / Streamlit) to hand off the pipeline to the frontend dashboard team.

Days 11–13: Refined GIS layer styling for direct dashboard embedding and exported map tiles/layers for handoff to the frontend team; worked with Aditya and Deepankar to embed the interactive GIS map into the live dashboard shell and tested rendering performance under the deployed environment.

Days 14–15: Fixed map projection and rendering issues identified during the testing phase, finalized GIS documentation, and supported the final dashboard demo by walking through the map layers and hotspot detection live.

Core Deliverables

- Mathematically sanitized dataset: fixed_atlantic_heatwaves.csv (3,243 clean, filtered spatial records).
- High-resolution spatial maps: vector print-layout charts isolating regional thermal plumes.
- Dashboard integration snippets: Python code blocks linking spatial data into interactive web maps.

Technologies Used

Python 3 (Pandas, NumPy); QGIS Desktop with EPSG:4326 / WGS 84 projections; Plotly Express rendered via the Streamlit web framework.

Key Contribution: *Solved the project's most technically tricky spatial-data bug (the 0° – 360° longitude wrap) early on, and carried that GIS expertise all the way through to a fully integrated, deployed map in the final dashboard.*

7.5 Shivansh Sharma – Database Engineer

Shivansh Sharma | Database Engineer | 15 Working Days

Roll No: 25SCS1003004651 | Aadhar No: 263943997350

Primary responsibility: Data Pipeline and Database Integration — managing the flow of processed data into the database, preparing it for backend integration, and supporting deployment.

Days 1–2: Took ownership of Data Pipeline and Database Integration during initial team meetings; analyzed the cleaned SST dataset from the collection team, mapping which columns were mission-critical for risk-prediction models and ensuring formatting was robust for database import.

Days 3–5: Built the storage foundation using PostgreSQL and pgAdmin; progressed from basic SQL

queries to managing complex tables; imported the satellite SST dataset and ran verification scripts, cross-referencing anomaly values and risk categories against source data for consistency and integrity.

Days 6–8: Developed a Python-based pipeline to automate CSV ingestion, connecting Python directly to PostgreSQL and testing for reliability; installed Node.js and configured the Prisma ORM; integrated PostGIS to handle spatial dimensions; synchronized the database with the shared Prisma schema, bridging raw data and the backend service.

Days 9–10: Worked with Sumit to refine data models and plan the Python-pipeline-to-live-database interaction; reviewed hardware-sensor research from Lakshya and began drafting a database structure for future real-time sensor data, noting that actual sensor hardware had not been procured for this project phase.

Day 11: Deployed the finalized database schema to a hosted PostgreSQL/PostGIS instance so the rest of the team could connect to a shared, always-available database.

Day 12: Optimized queries used for zone-risk lookups and alert generation, adding indexes on frequently queried spatial and time columns to improve response speed.

Day 13: Coordinated closely with Sumit on backend deployment, configuring secure database connection settings and access credentials for the live API.

Day 14: Conducted database load testing alongside the team's performance testing effort, confirming stable query response times under larger simulated datasets.

Day 15: Finalized database documentation, set up basic backup procedures, and supported the team's final demo and report review.

Technologies Used

PostgreSQL, pgAdmin, Python, SQL, Prisma ORM, Node.js, PostGIS.

Key Contribution: *Designed, built, and deployed the database backbone that every risk, alert, and dashboard query in the platform relies on — and stayed closely coordinated with the backend team through deployment.*

7.6 Sumit Sharma – Backend Developer & Deployment

Sumit Sharma | Backend Developer & Deployment | 15 Working Days

Roll No: 25SCS1003002486 | Aadhar No: 898424069533

Role: Creates API endpoints, data ingestion, alerts, and simulation services for the platform, and led the deployment of the backend system, working closely with the database and frontend teams.

Day 1: Reviewed the finalized project scope and data pipeline design with the team; planned the backend API architecture and drafted the initial list of endpoints needed for SST data, zone risk, reports, alerts, and simulation.

Day 2: Set up the Node.js backend project structure and environment configuration; defined the REST API route skeleton for SST records, coastal zones, and community reports.

Day 3: Implemented the SST data ingestion endpoint, working with the cleaned dataset supplied by the data team and validating incoming records before storage.

Day 4: Implemented the anomaly and risk-classification endpoint, connecting it to the geospatial database schema designed by Shivansh, so zone risk levels could be queried by the frontend.

Day 5: Built the community report submission and verification endpoints, including basic input validation and a check against risky-zone polygons to mark reports as verified or pending.

Day 6: Implemented the alert-generation service, which triggers automatically when a zone's risk score is classified as high or critical, and exposed an endpoint for the frontend to retrieve active alerts.

Day 7: Developed the simulation service endpoint that accepts user-defined simulation parameters (e.g., spread radius, duration) and returns predicted impact polygons for rendering on the map.

Day 8: Integrated the backend with the PostGIS/Prisma schema and tested API responses against database records to confirm consistency between stored data and API output.

Day 9: Conducted unit and integration testing across all endpoints, fixed edge cases (missing fields, invalid coordinates, malformed simulation parameters), and coordinated with Aditya to finalize endpoint contracts.

Day 10: Reviewed the complete backend module end to end with the team ahead of the integration and deployment phase, and finalized API documentation.

Day 11: Began deployment planning: selected a hosting platform for the backend API, and configured environment variables and secrets for a production-style setup.

Day 12: Deployed the backend API to the hosting service and connected it to the live PostgreSQL/PostGIS database, working closely with Shivansh to finalize the connection configuration.

Day 13: Conducted post-deployment smoke testing of all API endpoints in the live environment, confirming SST ingestion, risk classification, alerts, and simulation all functioned correctly once deployed.

Day 14: Resolved deployment issues (CORS configuration, environment variables) that surfaced during frontend integration, and supported Aditya in connecting the dashboard to the deployed API.

Day 15: Carried out final deployment verification across all endpoints, supported the team's demo rehearsal, and helped prepare the backend and deployment sections of the final report.

Technologies Used

Node.js, REST API design, PostgreSQL/PostGIS, Prisma ORM, Python (for data-ingestion coordination), cloud deployment/hosting configuration.

Key Contribution: *Not only built every core backend service the platform runs on, but also took the lead on deploying the backend to a live environment – turning the team's local work into an actually running, accessible system.*



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7.7 Aditya Pratap Singh – Frontend Developer

Aditya Pratap Singh | *Frontend Developer* | 15 Working Days

Roll No: 25SCS1003000571 | Aadhar No: 806024918757

Role: Builds the dashboard pages and responsive layout, translating the GIS, alerts, and reporting modules into an interactive web interface for NGO and end users, in close design collaboration with Deepankar.

Day 1: Reviewed the finalized project scope and deliverables list with the team; set up the frontend project structure and planned the main dashboard pages (overview, GIS map, alerts, community reports, admin panel).

Day 2: Researched UI patterns from existing NGO and environmental-monitoring dashboards; sketched low-fidelity wireframes for the main dashboard, GIS map page, and alerts page, reviewing early drafts with Deepankar.

Day 3: Implemented the base dashboard layout – navigation bar, sidebar, KPI summary cards, and a responsive grid system to support different screen sizes.

Day 4: Integrated an initial GIS map component into the dashboard shell and connected it to placeholder SST anomaly data supplied by the GIS and data teams.

Day 5: Built the SST Anomaly Visualizer page, including a colour-coded anomaly legend and layer-toggle controls for switching between SST, anomaly, and risk-zone layers.

Day 6: Developed the Community Reporting form UI (map-based location picker, incident details, input validation) and an initial layout for the admin moderation panel.

Day 7: Worked with Deepankar on refining the dashboard's colour palette, typography, and overall visual identity, and implemented loading and error states for the API-driven pages.

Day 8: Implemented the Alerts & Notifications panel along with trend charts and timeline widgets (using Plotly/Recharts-style components) to visualize risk history; collaborated with Deepankar on KPI card styling.

Day 9: Carried out responsive design testing across desktop and mobile breakpoints, refined the dark/light theme with Deepankar, and fixed early layout issues.

Day 10: Reviewed the complete frontend build end to end with Deepankar for UI consistency ahead of backend integration and deployment.

Day 11: Began integrating the live, deployed backend API into all dashboard pages, replacing placeholder data with real SST, anomaly, zone-risk, and alert data.

Day 12: Worked closely with Deepankar on final visual refinements as live data changed page layouts – adjusting spacing, card sizing, and chart scaling for real data ranges.

Day 13: Conducted cross-browser and cross-device testing on the deployed dashboard, and



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coordinated with Sumit to resolve API connectivity issues (CORS, endpoint URLs) surfaced during integration.

Day 14: Fixed UI bugs identified during the testing phase and polished the GIS map, alerts panel, and community reporting flow with input from Ayush and Deepankar.

Day 15: Finalized the deployed frontend build, and prepared and rehearsed the dashboard demo walkthrough for the final presentation.

Technologies Used

Streamlit / React-style dashboard components, Plotly Express for charts, responsive CSS layout, GIS map embedding, REST API integration.

Key Contribution: *Turned every other member's backend, data, and GIS work into a single usable interface – and, working hand-in-hand with Deepankar on design, delivered a dashboard that was both functional and genuinely presentable by demo day.*

7.8 Lakshya Narula – Deployment & Moderation, Hardware Research & Algorithm Support

Lakshya Narula | Deployment & Moderation / Hardware Research / Algorithm & Presentation Support | 15 Working Days

Roll No: 25SCS1003000518 | Aadhar No: 688239078650

Roles: Hardware Research, Algorithm Design Support, Presentation Preparation, and Deployment & Moderation (community reporting forms, admin review flow).

Day 1: Brainstormed the project idea, analyzed requirements, and accepted hardware-research, algorithm-support, and deployment & moderation responsibilities.

Day 2: Researched community-reporting and admin-moderation workflows used by similar NGO/environmental platforms to inform the design of the reporting and review flow.

Day 3: Outlined early requirements for the community reporting form and the admin moderation review flow, noting the fields and checks the moderation panel would need.

Day 4: Researched the ESP32 microcontroller and water-quality sensors (pH, turbidity, TDS, temperature) and documented hardware integration concepts as a planning exercise, since the physical components were not available to the team.

Day 5: Worked alongside Deepankar on the algorithm workflow and complexity analysis, helping structure it for the project's technical documentation.

Day 6: Prepared architecture diagrams and presentation slides aimed at an NGO audience.

Day 7: Refined the community reporting form requirements and the admin moderation review flow ahead of implementation.

Day 8: Prepared testing screens and satellite-data visualization concepts to support the testing module.

Day 9: Completed documentation for the moderation and hardware-research workstreams and coordinated handoff notes with the backend team.

Day 10: Presented the algorithm workflow and hardware research to the team; proposed a phased plan for future physical-sensor integration.

Day 11: Finalized the hardware research findings, documenting clearly that the ESP32 and sensor hardware could not be procured within the fifteen-day project timeline, and reframed the hardware integration plan as a recommended future phase rather than a built feature.

Day 12: Supported implementation of the admin moderation panel and reviewed the community-report moderation flow together with the backend team.

Day 13: Assisted with deployment-related testing tasks and cross-checked the moderation panel's functionality after the backend and dashboard were deployed.

Day 14: Helped update the presentation deck to accurately reflect the platform's final satellite-data-only scope, clearly noting the hardware extension as future work.

Day 15: Participated in the final demo rehearsal, presenting the moderation panel and the hardware research findings and future roadmap.

Skills Applied

Hardware research, algorithm design support, documentation, presentation skills, moderation-workflow design, and team collaboration.

Key Contribution: *Contributed across hardware research, algorithm workflow support, the moderation panel, and presentation preparation – while honestly documenting the hardware limitation so the final report and demo stayed accurate.*



8. Testing & Quality Assurance Summary

A dedicated Testing Module validated the complete software pipeline using satellite and historical water-quality/ocean datasets, covering data integrity, AI prediction accuracy, database reliability, dashboard performance, and user experience — extended in Days 11–15 to include the deployed, live environment. The system's Software Development Life Cycle (SDLC) followed six phases: Requirement Analysis, Design, Development, Testing, Deployment, and Maintenance.

8.1 Testing Scope

Testing and related activities (deployment, maintenance, updates) were estimated to account for 60–70% of total project effort, versus 30–40% for coding and implementation, underscoring the importance placed on quality assurance.

Functional Testing

Verified that dataset upload, dataset validation, preprocessing, AI prediction generation, dashboard visualization, alert generation, database storage, and historical-data retrieval each performed according to requirements — including alert generation when the Water Quality Index (WQI) falls below threshold or a prediction is classified Unsafe.

Non-Functional Testing

Evaluated performance, security, reliability, scalability, usability, and availability, confirming the application remains fast, secure, stable, and easy to use — including after deployment to a live hosting environment.

8.2 Additional Testing Performed

- Dataset validation testing — missing values, duplicates, invalid coordinates/timestamps/units.
- Data preprocessing testing — deduplication, missing-value handling, normalization, outlier removal.
- Unit testing — dataset validation, preprocessing, database, AI prediction, dashboard, and alert modules tested independently.
- Database testing — verified storage/retrieval of latitude, longitude, date, WQI, and AI prediction with no duplicate records, including on the deployed instance.
- AI model testing — accuracy, precision, recall, and F1-score evaluated against labelled data, targeting >90% accuracy.
- Verification & validation — confirmed the software matches design specifications and satisfies NGO/user requirements.
- Regression testing — confirmed new features (model updates, database changes, new datasets) do not break existing functionality.
- Smoke testing — confirmed critical functions (launch, upload, dashboard load, predictions, alerts) work immediately after deployment.



- Sanity testing – confirmed minor changes (layout tweaks, bug fixes, query optimization) work correctly without new issues.
- Performance / load & stress testing – measured loading time and response under normal and heavy workloads with large datasets, on both local and deployed environments.
- Alpha testing – internal team testing of AI accuracy, database, dashboard, and alerts.
- Beta testing – NGO representatives and potential end users evaluated usability, prediction usefulness, and alert clarity; feedback was incorporated into the final system.
- User Acceptance Testing (UAT) – NGO users evaluated usability, readability, and usefulness of the dashboard.

8.3 Cost of Bugs by Development Stage

Development Stage	Relative Bug-Fix Cost
Requirement Analysis	1×
System Design	5×
Development	10×
Testing	50×
Production	100×

8.4 Representative Test Cases

ID	Scenario	Expected Result	Status
TC-01	Missing values	Handled in preprocessing	Pass
TC-02	Duplicate records	Removed	Pass
TC-03	Invalid coordinates	Flagged	Pass
TC-04	AI classification	Correct prediction	Pass
TC-05	Dashboard with 100k rows	Loads successfully	Pass
TC-06	Unsafe prediction	Alert generated	Pass
TC-07	Deployed API smoke test	All endpoints respond correctly	Pass

8.5 Summary

Testing demonstrated that the system reliably processes satellite-derived environmental datasets, cleans and validates the data, generates accurate AI predictions, stores results in the database, visualizes trends through the dashboard, and issues timely alerts – end to end, in a fully deployed environment. The prototype achieved high prediction accuracy (>90%), fast response times, and stable operation with large datasets, making it suitable for NGO pilot studies and environmental decision support.

9. Final Deliverables

- Deployed, interactive dashboard with overview, GIS map, alerts, and reporting pages.
- GIS map with SST, anomaly, and risk-zone layers.
- SST anomaly visualizer with colour-coded legend and layer toggles.
- Community reporting module with map-based submission and admin verification.
- Simulation module for spread/heatwave impact prediction.
- Admin moderation panel for reviewing community reports.
- Deployed backend API and PostgreSQL/PostGIS database with automated ingestion pipeline.
- Complete Testing Module report validating the pipeline end to end, including the deployed environment.
- Documented hardware-extension plan (ESP32 + sensors) as a recommended future phase.
- Final report, presentation (PPT), and demo video.

10. Challenges & Solutions

Area	Challenge	Solution
Data	Inconsistent longitude conventions (0°–360° vs –180°/+180°) across datasets	Standardized coordinates early and verified conversions against known reference points
Data	Risk of over-cleaning near coastal/ice-covered and missing-data regions	Cleaned incrementally with statistics checked after every step; re-verified flagged rows as a team
GIS	Choosing an appropriate spatial aggregation level; achieving a clear, accurate visualization	Validated aggregated (10° grid) patterns against original high-resolution data; iteratively tuned QGIS classification and IDW interpolation settings
Analysis	Integrating SST anomaly with ocean chemistry into one risk framework	Grounded the multi-parameter risk model only in relationships consistently supported by scientific literature
Database/Backend	Bridging a Python-based data pipeline with a Node.js/Prisma backend and PostGIS schema, then deploying it	Automated CSV ingestion in Python, synchronized schemas via Prisma, cross-verified imported records, and deployed the database and API to a hosted environment
Frontend	Translating GIS, alert, and reporting modules into a coherent, responsive, deployed UI	Iterative wireframing, component-based layout, close Deepankar–Aditya design collaboration, and usability refinement based on Testing Module feedback



Area	Challenge	Solution
Hardware	ESP32 microcontroller and sensors were not procured or available during the project period	Documented the hardware research and integration plan as a future-phase recommendation, and delivered a fully software/satellite-data-based platform for this phase
Team	Coordinating eight members across data, GIS, backend, database, frontend, deployment, and documentation workstreams over fifteen days	Daily role-based updates, a shared roadmap, close cross-functional pairing at integration points, and a documentation lead consolidating progress throughout

11. Conclusion

Over fifteen working days, the team progressed from an initial concept for a low-cost, sensor-based water quality monitoring system to a fully scoped and deployed geospatial marine intelligence platform combining satellite SST monitoring, GIS visualization, ocean chemistry analysis, community reporting, and simulation. Each member contributed a distinct, well-documented piece of the pipeline – from raw data collection and cleaning, through GIS correction and visualization, database and backend development and deployment, frontend and UI design, to algorithm design and rigorous testing – resulting in a coherent, scalable, and thoroughly tested platform.

While the ESP32 hardware and sensor kit planned at project initiation were not available during this phase, the team adapted successfully, delivering a complete satellite-data-driven platform and leaving hardware integration as a clearly documented next step. Every member gave their best effort across all fifteen days, and the team's close, consistent coordination – daily updates, shared planning, and cross-functional collaboration between data, GIS, backend, database, frontend, and documentation – is what allowed eight individual contributions to come together into one working, deployed system.

The project demonstrates how theoretical concepts (climatology baselines, coordinate systems, algorithmic complexity, and software testing principles) translate directly into a real, end-to-end data platform, and leaves the team with a strong foundation and clear roadmap for continued development, including future physical sensor integration (ESP32) and expanded NGO deployment.

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